

Lecturer Student Progress Tracking Web App

Project brief, system concept, dashboard requirements, milestone templates, and development instructions

Purpose	To help lecturers track FYP, Master, and PhD student progress using milestones, weekly tasks, meeting summaries, deadlines, progress bars, and priority ranking.
Main users	Lecturer, FYP Student, Master Student, PhD Student
Core idea	A supervision control center that shows who is on track, who is delayed, what task is urgent, and who is close to graduation.
Output	A web application with role-based dashboards, progress monitoring, deadline calendar, publication/thesis tracker, and automatic student ranking.

Prepared as an initial software requirement and design draft. This document can be used as a guide before building the prototype or explaining the idea to a developer/team.

1. Project Overview

The proposed web app is designed for university lecturers who supervise students from different study categories. Each category has a different timeline, milestone structure, and completion requirement. The app should reduce dependence on Excel sheets, WhatsApp messages, and manual memory by placing progress updates, meeting notes, deadlines, and research outputs in one system.

The system must allow students to update weekly progress while allowing the lecturer to see the overall supervision situation from a single dashboard.

Main goals

- Track FYP, Master, and PhD students using different milestone templates.
- Allow students to submit weekly progress, blockers, evidence, and meeting summaries.
- Show progress bars for research achievement and time-to-graduation.
- Help lecturers identify urgent students and urgent tasks based on deadlines and risk.
- Rank students by attention priority and fast-completion potential.
- Provide clear progress reports that can be exported later as PDF or Excel.

2. User Roles

Role	Main responsibility	Dashboard focus
Lecturer	Create students, assign templates, review updates, give feedback, approve tasks, manage deadlines, monitor risk.	All students, calendar, priority ranking, task urgency, publication risk, thesis progress.
FYP Student	Update weekly tasks, upload draft chapters, report meeting summaries, complete semester milestones.	Semester 1 and Semester 2 milestones, chapter progress, final report readiness.
Master Student	Update objective progress, thesis progress, one publication requirement, weekly tasks, and meeting summaries.	Objective 1-3, Paper 1, thesis chapters, submission target.
PhD Student	Update objective progress, two-paper publication requirement, thesis progress, weekly tasks, and meeting summaries.	Research objectives, Paper 1, Paper 2, thesis, viva/submission readiness.

3. Student Milestone Templates

Each student type must use a different milestone template. The lecturer should be able to customize these templates later, but the first version can use default templates.

3.1 FYP Milestone Template

Period	Milestone	Expected output
Semester 1	Chapter 1 - Introduction	Problem statement, research questions, scope, and clear objectives.
Semester 1	Chapter 2 - Literature Review	Related articles reviewed and research gap identified.
Semester 1	Chapter 3 - Methodology	Method, tools, data, experiment design, and planned analysis.
End of Semester 1	Initial result	At least one result with early discussion.
Semester 2	Continue result/analysis	More complete experiment or analysis based on Semester 1 result.

Period	Milestone	Expected output
Semester 2	Chapter 4 - Results and Discussion	Complete results, figures, tables, and discussion.
Semester 2	Chapter 5 - Conclusion	Conclusion, contribution, limitation, and future work.
End of Semester 2	Final submission	Final thesis/report submitted and corrected.

3.2 Master Milestone Template

Component	Requirement	Tracking details
Duration	Usually around 2 years; maximum around 3 years.	System tracks time used against expected progress.
Research objectives	Usually 3 objectives.	Each objective should have tasks, deadline, evidence, and completion status.
Publication	1 article required.	Track idea, result, draft, supervisor review, submission, revision, acceptance/publication.
Thesis	Thesis must be submitted.	Track Chapter 1-5, full draft, correction, submission, viva/correction if applicable.
Graduation readiness	Based on objective completion, paper status, thesis completion, and remaining time.	Shown using progress bars and risk status.

3.3 PhD Milestone Template

Component	Requirement	Tracking details
Duration	Fast completion around 3 years; maximum around 5 years.	System tracks expected progress versus time used.
Research objectives	Multiple objectives, usually at least 3.	Each objective linked to tasks, publications, and thesis chapters.
Publication	2 articles required.	Track Paper 1 and Paper 2 separately from draft to accepted/published.
Thesis	Full PhD thesis must be submitted.	Track proposal, chapters, integration, full draft, notice of submission, viva, correction.
Graduation readiness	Based on objectives, two papers, thesis, viva/submission status, and remaining time.	Shown using progress bars and risk indicators.

4. Weekly Task and Meeting Update System

The weekly task system is the core activity flow. Students must update what they have completed, what is still pending, what problem they faced, and what they plan to do next. This allows the lecturer to know the expectation for next week and compare planned work against actual progress.

Field	Description	Example
Week/date	The reporting week and update date.	Week 5, 20 May 2026
Completed this week	Work finished during the week.	Completed Chapter 3 methodology draft.
Not completed	Planned work that was not finished.	Result table still pending.
Blocker/problem	Issue stopping progress.	Dataset cleaning not complete.
Plan for next week	Expected work before the next update or meeting.	Run experiment and prepare first result figure.
Related milestone	Milestone linked to the weekly task.	Objective 1 / Chapter 4 / Paper 1
Deadline	Target completion date.	27 May 2026
Evidence upload	File proof of progress.	Word draft, PDF paper, Excel result, image, slide.
Need feedback	Whether supervisor review is needed.	Yes - check methodology flow.

Meeting summary module

Field	Purpose
Meeting date	Records when the supervision meeting happened.
Discussion summary	Summarizes what was discussed.
Supervisor feedback	Stores lecturer comments and requested changes.
Action items	Lists what the student must complete next.
Deadline	Connects action items to a due date.
Next meeting	Schedules or suggests the next discussion date.

5. Lecturer Dashboard Requirements

The lecturer dashboard should work like a supervision control center. It must show multiple sections, not only a simple student list.

Dashboard section	What it shows	Why it is useful
Overview cards	Total students, FYP count, Master count, PhD count, delayed students, tasks due this week, overdue tasks, near-graduation students.	Gives a fast summary of the supervision load.
Calendar view	Meetings, deadlines, thesis submissions, paper revision deadlines, FYP presentation dates.	Helps lecturer plan weekly attention.
Progress overview	Progress bars for each student and comparison between progress achieved and time used.	Shows whether student progress is reasonable for their study duration.
Priority student ranking	Students sorted by urgency and risk.	Tells lecturer who needs attention first.
Task priority list	Urgent tasks across all students sorted by deadline and importance.	Tells lecturer what to check first.
Weekly update monitor	Who submitted this week's update and who has been inactive.	Prevents students from disappearing for weeks.
Meeting summary	Latest meeting notes and next actions.	Helps lecturer remember agreed tasks.

Dashboard section	What it shows	Why it is useful
Publication tracker	Master Paper 1, PhD Paper 1 and Paper 2 status.	Highlights publication risk.
Thesis tracker	Chapter progress, full draft, submission, viva/correction status.	Shows writing progress clearly.
Fast completion candidates	Students likely to finish soon with focused support.	Helps lecturer push near-completion students to graduate faster.

6. Priority and Ranking Logic

The app should not only store progress. It should also help the lecturer decide where to focus. Before using the dashboard, the lecturer can choose what kind of task or student should be prioritized.

Lecturer priority setup

Priority option	Meaning
Deadline urgency	Focus on students with near or overdue deadlines.
Graduation risk	Focus on students whose progress is behind their time used.
Publication progress	Focus on Master/PhD students with delayed paper requirement.
Thesis writing	Focus on students with delayed thesis chapters.
Weekly activity	Focus on students who have not updated recently.
Fast completion potential	Focus on students who can finish quickly with support.
Low-performing students	Focus on students with repeated delays.
High-performing students	Focus on students who are close to completion and can be pushed to graduate.

Suggested scoring model

Factor	Example scoring idea
Overdue deadline	+30 points if overdue; +15 if due within 7 days.
No weekly update	+20 points if no update for 2 weeks; +10 if no update this week.
Progress behind time	+25 points if time used is much higher than milestone progress.
Publication risk	+25 points if Master/PhD paper is not drafted/submitted by expected stage.
Thesis delay	+20 points if key chapter is late.
Near graduation	+15 points if student is above 75% complete but has remaining critical task.
Repeated missed deadlines	+10 to +20 points based on number of missed deadlines.

The system should produce two rankings

Ranking	Question answered	Students shown
Attention Priority Ranking	Who needs my help urgently?	Delayed, inactive, overdue, publication-risk, or critical students.
Fast Completion Ranking	Who can finish soon if I support them now?	Students with high progress, good consistency, and only a few critical tasks remaining.

7. Progress Bar and Graduation Readiness

Students should see how much time they have left and how close they are to graduation. The lecturer should also see whether the student progress is healthy compared with the time already used.

Progress bar	Purpose	Example
Overall graduation progress	Combines milestone, objective, publication, and thesis completion.	Master student: 55% complete.
Time used	Shows how much of the maximum study duration has been used.	PhD student: 2.5 years used out of 5 years = 50%.

Progress bar	Purpose	Example
Objective progress	Shows completion for each research objective.	Objective 1: 100%, Objective 2: 60%, Objective 3: 20%.
Publication progress	Shows paper stage from idea to acceptance.	Paper 1 submitted = 70%; accepted = 100%.
Thesis progress	Shows chapter and submission progress.	Chapter 1-3 done, Chapter 4 draft, Chapter 5 pending.

Example overall progress weighting

Student type	Suggested weighting
FYP	Chapter/milestone progress 80%, final submission/presentation 20%.
Master	Objectives 40%, Paper 1 25%, thesis 25%, submission/viva/correction 10%.
PhD	Objectives 35%, Paper 1 20%, Paper 2 20%, thesis 20%, submission/viva/correction 5%.

8. Recommended Screens

The first prototype should include these main pages. More advanced analytics can be added after the core workflow is stable.

Side	Screen	Purpose
Lecturer	Dashboard	Overview cards, calendar, priority ranking, task list, weekly update status.
Lecturer	Students	Add, edit, filter, and open student profiles.
Lecturer	Student Profile	Detailed progress, milestones, weekly history, files, feedback.
Lecturer	Milestone Templates	Configure FYP, Master, and PhD milestone templates.
Lecturer	Reports	Export progress report by student or by cohort.
Student	My Progress	Personal progress bars, current tasks, deadlines, supervisor feedback.
Student	Weekly Update	Submit completed work, pending work, blockers, next-week plan.
Student	My Milestones	View required milestones and completion status.
Student	Meeting Summary	Submit or review supervision meeting notes.
Student	Evidence Upload	Upload draft thesis, paper, result table, slide, or image.

9. Suggested Database Structure

This is a simplified database design for the development team. The final design can be adjusted depending on the framework used.

Table	Important fields	Purpose
users	id, name, email, password_hash, role	Stores login accounts for lecturer and students.
students	id, user_id, lecturer_id, student_type, start_date, expected_end_date, max_end_date, research_title	Stores student profile and study category.
milestone_templates	id, student_type, title, description, default_order	Stores default FYP/Master/PhD templates.
student_milestones	id, student_id, template_id, title, deadline, status, progress_percent	Stores assigned milestones for each student.
objectives	id, student_id, objective_no, description, progress_percent, status	Tracks Master/PhD research objectives.
publications	id, student_id, paper_no, title, target_journal, status, submission_date, acceptance_date	Tracks Master/PhD paper requirements.
weekly_updates	id, student_id, week_start, completed, not_completed, blocker, next_plan, related_milestone_id	Stores weekly progress updates.
meeting_logs	id, student_id, meeting_date, summary, supervisor_feedback, action_items, next_meeting	Stores meeting summaries and agreed actions.
tasks	id, student_id, title, description, priority, deadline, status, related_area	Stores detailed tasks and deadlines.
feedback	id, lecturer_id, student_id, item_type, item_id, comment, created_at	Stores lecturer comments on tasks, updates, files, or milestones.
files	id, student_id, uploaded_by, file_url, file_type, related_item	Stores uploaded evidence.

10. Development Instructions

These instructions can be given to a developer or used as the starting point for building the prototype.

10.1 Build order

Phase	What to build	Minimum requirement
Phase 1	Authentication and role-based dashboard	Lecturer, FYP Student, Master Student, PhD Student login and redirects.
Phase 2	Student management	Lecturer can add student, choose type, set start date, and assign template.
Phase 3	Milestone tracker	FYP, Master, and PhD default milestone templates with status and progress.
Phase 4	Weekly update and meeting summary	Students submit updates; lecturer reviews and comments.
Phase 5	Dashboard analytics	Overview cards, calendar, progress overview, overdue tasks, weekly update monitor.
Phase 6	Priority ranking	Attention priority ranking and fast-completion ranking.
Phase 7	File uploads and reports	Upload evidence and export progress report.

10.2 Important business rules

- Every student must have one student type: FYP, Master, or PhD.
- Student type determines the default milestone template.
- FYP progress is mainly semester and chapter based.
- Master progress must include objectives, one publication, and thesis submission.
- PhD progress must include objectives, two publications, and thesis submission.
- Weekly updates should be required or strongly reminded every week.
- A task becomes overdue when its deadline has passed and status is not completed or approved.
- Progress submitted by student can be marked as pending until lecturer approves it.
- Dashboard ranking should update automatically when tasks, deadlines, weekly updates, or progress status changes.
- Progress percentage should not only depend on student self-declaration; important tasks should require lecturer endorsement.

10.3 Recommended status labels

Area	Suggested labels
Task status	Not started, In progress, Submitted, Need revision, Approved, Completed, Overdue.
Student risk	On track, Slight delay, Delayed, Critical, Completed.
Publication status	Idea, Result ready, Drafting, Supervisor review, Submitted, Under review, Revision, Accepted, Published.
Thesis status	Not started, Drafting, Submitted to supervisor, Need revision, Approved, Submitted to faculty.

11. Example Lecturer Dashboard Layout

Dashboard row	Content
Top row	Summary cards: Total students, Critical students, Tasks due this week, Overdue tasks, Near graduation.
Middle row	Left: Calendar. Right: Priority student ranking.
Next row	Left: Progress overview. Right: Weekly update monitor.
Next row	Left: Task priority list. Right: Recent meeting summaries.
Bottom row	Publication tracker, thesis tracker, and fast completion candidates.

12. Final Concept Summary

This system should be designed as a smart supervision dashboard. The student side focuses on weekly reporting, milestone progress, meeting summaries, and evidence uploads. The lecturer side focuses on monitoring, prioritizing, feedback, progress approval, deadlines, and graduation readiness.

In simple terms, the app should answer these questions for the lecturer:

- Who is delayed?
- Who has a deadline this week?
- Who did not update their weekly progress?
- Which task should be checked first?
- Which student is close to graduation?
- Which Master or PhD student has publication risk?
- What was discussed in the last meeting?
- What should each student prioritize next week?

The most valuable feature is the combination of milestone tracking, weekly updates, deadline calendar, priority ranking, and graduation progress bar in one platform.